FORMATIVE ASSESSMENT – 4 (FA-4)

MATHEMATICS • CLASS 9 • SLIP TEST

Question 1

Three angles of a quadrilateral are **75°**, **90°**, and **75°**. Find the measure of the fourth angle.

Question 2

In a parallelogram **ABCD**, the adjacent angles **∠A** and **∠B** are in the ratio **2 : 3**. Find the measures of all four angles of the parallelogram.

Question 3

A chord of a circle of radius **10 cm** is at a distance of **8 cm** from the centre. Calculate the length of the chord.

Question 4

In a cyclic quadrilateral **ABCD**, if opposite angle **∠A = 70°** and adjacent angle **∠B = 100°**, determine the measures of **∠C** and **∠D**.

Question 5

Find the mean of the first six prime numbers.

SOLUTIONS & SCHEME OF WORK

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Solution 1

The sum of all four interior angles of a quadrilateral is **360°**:

$75° + 90° + 75° + x = 360°$

$240° + x = 360° \Rightarrow x = 360° - 240° =$ **120°**

Ans: Fourth angle = 120°

Solution 2

Adjacent angles of a parallelogram are supplementary (**∠A + ∠B = 180°**):

Let $\angle A = 2x$ and $\angle B = 3x$. Then $2x + 3x = 180° \Rightarrow 5x = 180° \Rightarrow$ **x = 36°**

- $\angle A = 2(36°) =$ **72°**, $\angle B = 3(36°) =$ **108°**

Opposite angles are equal: $\angle C = \angle A =$ **72°**, $\angle D = \angle B =$ **108°**

Ans: ∠A = 72°, ∠B = 108°, ∠C = 72°, ∠D = 108°

Solution 3

The perpendicular drawn from the centre to a chord bisects the chord:

Let radius r = 10 cm, perpendicular distance d = 8 cm, half-chord = L.

By Pythagoras Theorem: $L^2 + d^2 = r^2 \Rightarrow L^2 + 8^2 = 10^2$

$L^2 + 64 = 100 \Rightarrow L^2 = 36 \Rightarrow$ **L = 6 cm**

Total length of chord = 2 × L = 2 × 6 = **12 cm**

Ans: Length of chord = 12 cm

Solution 4

In a cyclic quadrilateral, the sum of either pair of opposite angles is **180°**:

- $\angle A + \angle C = 180° \Rightarrow 70° + \angle C = 180° \Rightarrow$ **∠C = 110°**
- $\angle B + \angle D = 180° \Rightarrow 100° + \angle D = 180° \Rightarrow$ **∠D = 80°**

Ans: ∠C = 110°, ∠D = 80°

Solution 5

The first six prime numbers are: **2, 3, 5, 7, 11, 13**

Sum of observations = 2 + 3 + 5 + 7 + 11 + 13 = **41**

Number of observations = 6

Mean = Sum / Number = 41 / 6 = **6.83** (or 6 5/6)

Ans: Mean = 41 / 6 ≈ 6.83